
SigIQ

Signal Analysis Workstation for Automated RF Signal Identification and Recovery

Technical Report: Architecture, Algorithms and Interface Walkthrough



Abstract

This report documents **SigIQ**, a Tkinter desktop application and numpy/scipy DSP engine that takes a raw, headerless `.iq` or `.wav` recording and determines, end to end and with no manual signal-processing work, its modulation, its real transmission parameters (sample rate, symbol rate, carrier offset, bandwidth, SNR), and its recoverable bitstream (de-interleaving, forward error correction, header/payload identification). Every candidate modulation is demodulated and scored against the data rather than assumed, every reported parameter carries a label for where it actually came from, and the system reports `ambiguous` or `insufficient_evidence` rather than forcing a confident-looking wrong answer. The report walks through the software's architecture and control flow, derives the governing signal-processing equations behind parameter estimation, evidence scoring and soft-decision decoding, and reproduces the interface exactly as generated by the live application.

1 Overview

A radio transmission is just a stream of bits encoded onto a wave. Unlike a `.jpg` or `.mp3`, there is no universal file format for it: the sender and receiver privately agree in advance on how fast bits are sent (the symbol rate), how 0s and 1s are represented as changes in the wave (the modulation), whether extra data is mixed in to survive noise (interleaving and forward error correction), and whether a fixed pattern marks where the real content starts (a header/sync word). Knowing that agreement is normally a precondition for reading the signal at all.

SigIQ works the other way around: given only a raw recording, with none of that context, it tests the reasonable possibilities for each of those unknowns and reports which combination the data actually supports, plus the recovered bits if enough of the signal survived.

Question	How it is answered
What modulation is being used?	Demodulate against every candidate (BPSK, QPSK, 16-QAM, 2-FSK, 4-FSK) and score which one the data actually supports best.
What are the real transmission parameters?	Sample rate, symbol rate, carrier offset, bandwidth and SNR, each labelled as user-provided, read from a file header, estimated, or inferred from the winning candidate.
Are the bits interleaved and/or FEC-coded?	A joint search over de-interleaving (block, convolutional, diagonal, pseudo-random) and FEC (Viterbi, Reed-Solomon, concatenated, LDPC), kept only if a header match or a verifying decoder supports it.
Is there a recognizable header/payload?	Correlation against known sync words (CCSDS ASM, HDLC flag, Barker-11), backed by a statistical false-alarm test, not a similarity guess.

Design principle: never hardcode a result

Every number shown in the interface is computed from the loaded signal, on that run, by the DSP engine; nothing is a fixed placeholder. Three concrete guarantees follow from that rule, each enforced by an automated test:

- A “decoded” result only appears if a decoder’s parity/syndrome check genuinely passes, or a Viterbi decode’s bit-mismatch rate is statistically far enough below what a random stream produces (Section ??).
- A “header found” result only appears if the correlation peak clears a binomial false-alarm test at $p < 0.01$, Bonferroni-corrected across every sync word tried (Section ??).
- Pure noise input is reported as **insufficient_evidence** with nothing recovered, verified by a test that feeds the pipeline seeded random noise and asserts nothing downstream fires.

2 Innovation: Evidence-Backed Hypothesis Validation

The preceding design principle (never report a result the evidence does not support) is not just a coding convention; it is the core architectural idea behind SigIQ, and it is worth stating precisely: *SigIQ does not classify an unknown radio signal and commit to that label. It generates competing signal hypotheses, processes each one through demodulation and, where applicable, de-interleaving and FEC, fuses the resulting evidence, rejects hypotheses that fail validation, and reports **insufficient_evidence** when no hypothesis has enough support to be named.*

2.1 Classification-first versus hypothesis-then-evidence

A conventional automatic modulation recognition tool commits early: a classifier reduces the signal to a feature vector, names the most likely modulation, and everything downstream (demodulation, FEC decoding) assumes that label is correct. If the classifier is wrong, every later stage inherits that error silently. Figure ?? contrasts that flow with SigIQ’s own.

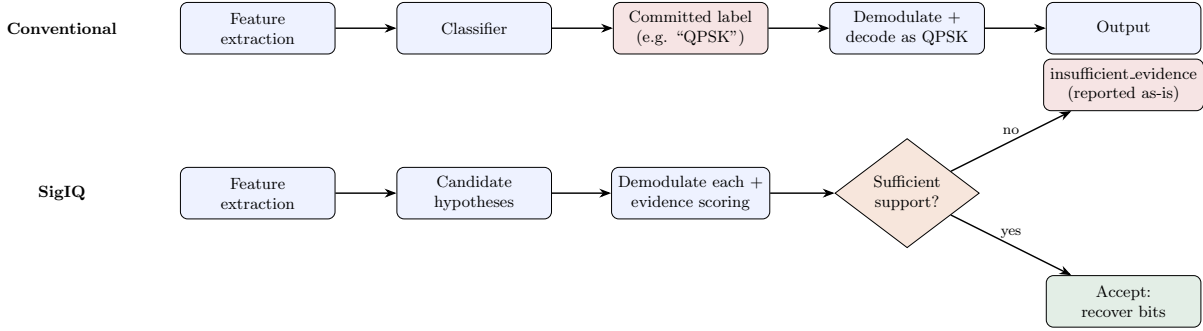


Figure 1: A conventional classify-then-process pipeline commits to one label before any downstream evidence exists. SigIQ scores every candidate (Section ??) and only proceeds to recovery if the winner clears the verdict thresholds (Figure ??); otherwise it reports `insufficient_evidence` rather than an unsupported guess.

2.2 Open-set rejection: a genuine “unknown” outcome

Because the verdict logic in Section ?? checks timing fit, best score and score margin *before* naming a winner, a signal with no real symbol clock, or whose best candidate simply does not fit well enough, is not forced into the nearest label. This is open-set rejection, not a hedge: it is exercised by `tests/test_behaviour.py::test_pure_noise_is_not_classified`, which feeds the pipeline seeded random noise and asserts the result is `insufficient_evidence` with nothing recovered, not a low-confidence guess at some modulation.

2.3 Cross-layer evidence into decoding

Evidence is not confined to the constellation. Section ??’s soft-decision LLRs carry the demodulator’s own per-bit confidence through de-interleaving into the FEC decoder, and Section ??’s significance test turns a raw decode into a statistical statement (“this stream decodes far better than chance”) rather than a bare success flag. This currently applies to Viterbi and LDPC, which have a natural soft-input form; Reed-Solomon’s classical Berlekamp-Massey decoding does not, and is used hard-decision only (Section ??). Rather than presenting every decoder as exposing one common probability, the evidence each one contributes is normalized separately into the same $[0, 1]$ -scored framework used for modulation candidates.

2.4 A canonical representation across acquisition formats

A headerless `.iq` recording and a `.wav` recording carry their sample data differently and, in the `.iq` case, no metadata at all. Both are read into the same `RawSignal` object (`core/pipeline/models.py`): complex baseband samples, sample rate, source format, center frequency and filename. Every stage from isolation onward operates on this one representation, so the rest of the pipeline does not need to know, or care, which file format a recording arrived in.

2.5 Toward joint modulation and coding validation

De-interleaving and FEC are already searched *jointly* with each other, not as a fixed chain (Section ??, point 5): a de-interleaver/FEC pair is only accepted if the combination shows real supporting evidence, header correlation or a genuinely verifying decode. Today that joint search runs on the bits of the already-selected best-supported modulation hypothesis. The natural extension of the same evidence-fusion principle, and a direction this architecture is built to support rather than a claim made about the current release, is to let that downstream recovery evidence feed back into modulation-hypothesis scoring itself, so a modulation whose bits also decode cleanly under some FEC is preferred over one that merely fits its constellation.

The scoring interface (Section ??) was deliberately kept as a normalized, per-candidate $[0, 1]$ metric set for exactly this reason: adding a fourth, downstream-recovery metric does not require changing how any existing metric is computed or combined.

3 Software Architecture

SigIQ follows a strict two-layer, one-direction architecture: control only ever flows downward, data only ever flows back up as a single result object, and the DSP engine has no idea a graphical interface exists.

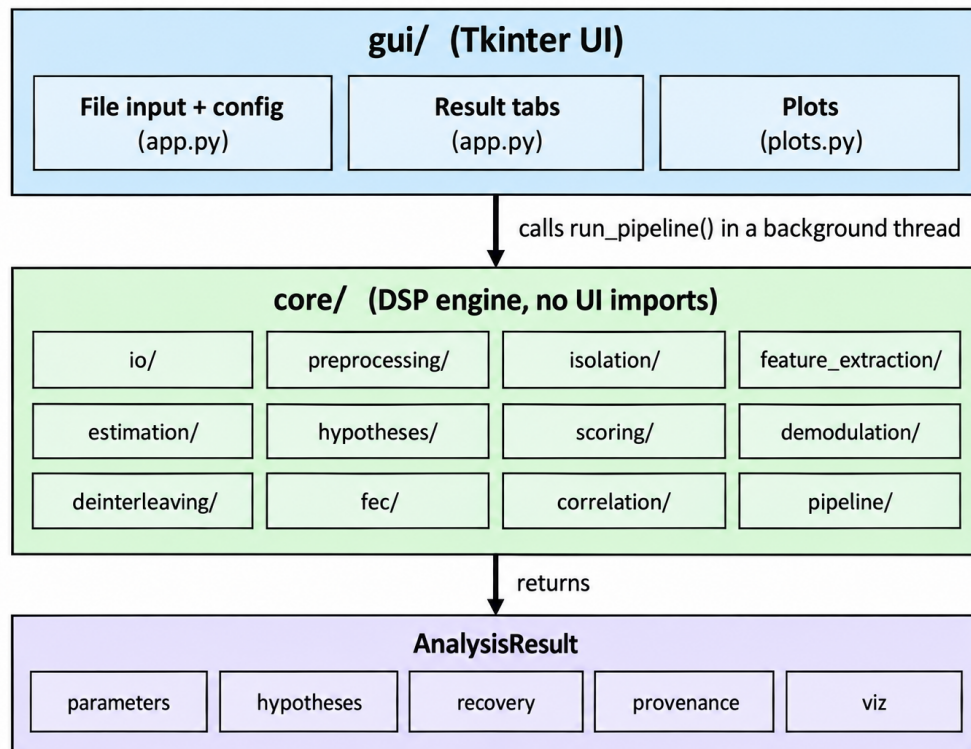


Figure 2: SigIQ architecture: the `gui/` layer calls `run_pipeline()` in a background thread; the `core/` DSP engine (twelve packages, no UI imports) returns a single **AnalysisResult** object that the GUI only renders.

Layer	Role
gui/ (Tkinter UI)	File input and configuration, the four result tabs, and the plots, all in <code>app.py</code> and <code>plots.py</code> . Its only job is to call <code>run_pipeline()</code> and render whatever comes back; it contains no signal-processing code of its own.
core/ (DSP engine)	Twelve packages covering the whole signal chain, from loading through the pipeline orchestrator itself: <code>io</code> , <code>preprocessing</code> , <code>isolation</code> , <code>feature_extraction</code> , <code>estimation</code> , <code>hypotheses</code> , <code>scoring</code> , <code>demodulation</code> , <code>deinterleaving</code> , <code>fec</code> , <code>correlation</code> , <code>pipeline</code> . None of them import anything from <code>gui/</code> , which is what makes <code>core/</code> runnable and testable completely on its own.

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Layer	Role
AnalysisResult	The one object <code>core/</code> hands back: estimated parameters, scored hypotheses, recovered bits, a provenance label for every value, and the visualization data the plots render.

4 Pipeline Flow

The pipeline is implemented in `core/pipeline/analyzer.py`.

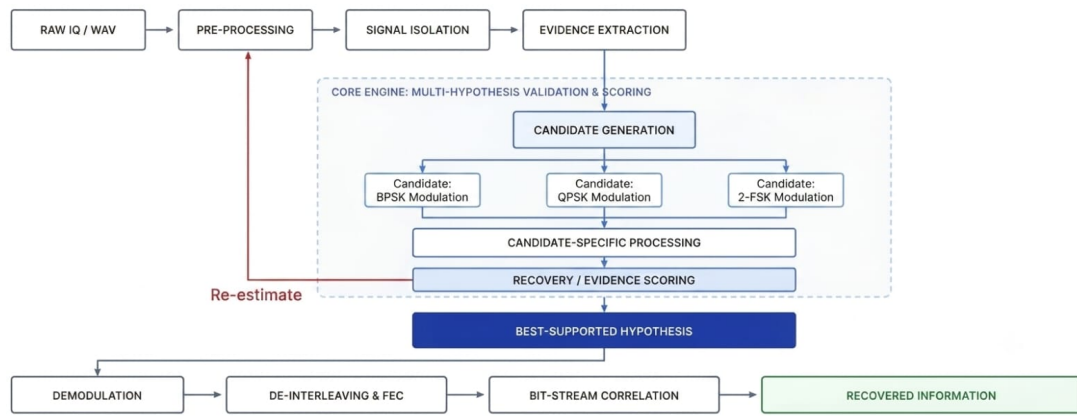


Figure 3: SigIQ pipeline: raw IQ/WAV through pre-processing, isolation and evidence extraction into the multi-hypothesis core engine, out to the best-supported hypothesis, and finally through de-interleaving, FEC and bit-stream correlation to the recovered information.

- Raw IQ/WAV → Pre-processing → Signal Isolation → Evidence Extraction.** The file is loaded, DC offset and power are normalized (with optional denoising), the active segment and channel are isolated, and spectral/temporal/statistical features plus initial parameter estimates are extracted.
- Core engine: multi-hypothesis validation and scoring.** Candidate modulations (BPSK, QPSK, 16-QAM, 2-FSK, 4-FSK by default) are generated in parallel. Each one is demodulated and scored on constellation fit, order consistency and timing fit *inside this box*, *not after it*: a candidate cannot be scored without being demodulated first (Section ??).
- Re-estimate.** If the result is weak or ambiguous, the search retries. Most retries are a bounded coordinate search over symbol rate, carrier offset and low-pass cutoff that loops straight back into evidence scoring using the samples already isolated; only once that search is exhausted and the result is still weak does a new preprocessing profile get tried from scratch (Figure ??).
- Best-Supported Hypothesis.** The winner's verdict is one of `determined`, `ambiguous`, `insufficient_evidence` or `user_selected` (Section ??), and its bits (already produced during scoring) carry forward.
- De-interleaving & FEC → Bit-stream Correlation → Recovered Information.** De-interleaving and FEC are searched jointly, ranked by header correlation and FEC success. Known sync-word matching then locates a header/payload region if the false-alarm test allows it.

4.1 Scoring and re-estimation in detail

Figure ??’s single “Re-estimate” arrow back to pre-processing is really the fallback of two nested retry loops.

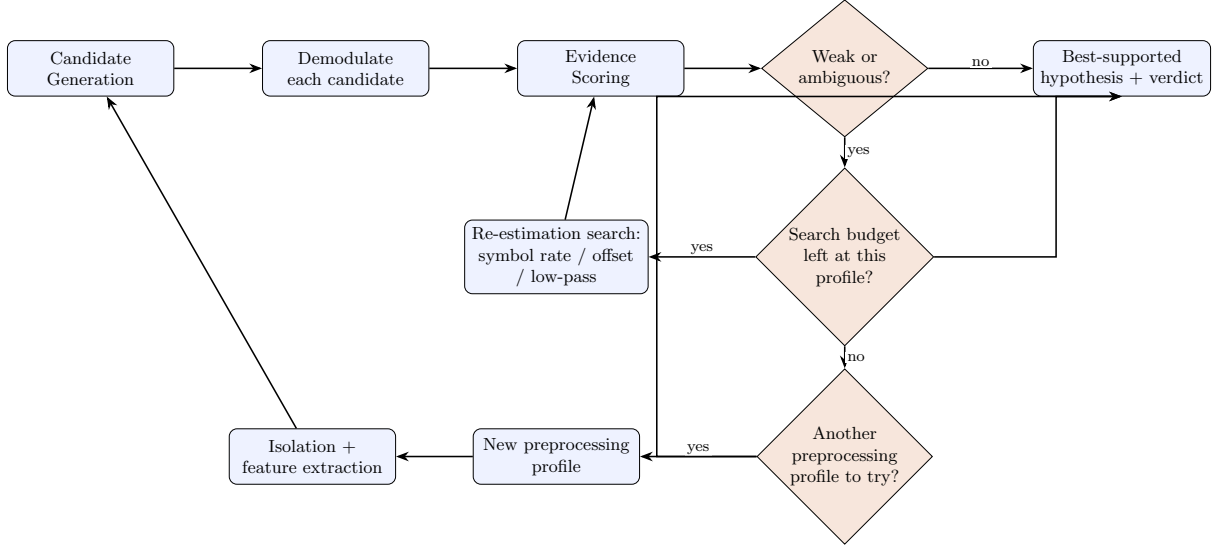


Figure 4: The inner loop (Re-estimation search → Evidence Scoring) fires on almost every weak/ambiguous result, re-scoring at nearby symbol-rate/offset/cutoff points without leaving the current preprocessing profile. The outer loop (New preprocessing profile → Isolation → Candidate Generation) is what Figure ??’s single arrow represents: it only runs once the inner search is exhausted and the result is still weak.

The inner search is a bounded coordinate search over symbol rate, carrier offset and low-pass cutoff, with objective equal to the best candidate’s evidence score. It stops when an iteration improves the score by less than a tolerance (0.005) or after a fixed iteration cap. A higher symbol-rate neighbour is only accepted on a tie if its clock-line support exceeds the current point’s by a real margin (`HARMONIC_TIE_MARGIN`, 10%), which prevents the search from jumping to a harmonic of the true rate on a noise-level difference.

5 Signal-Processing Algorithms

5.1 Demodulation and constellation fit

PSK candidates of order M use the ideal unit-circle constellation $c_k = e^{j2\pi k/M}$, $k = 0, \dots, M-1$; QAM candidates of order M use a square grid normalized to unit average power. Received symbols are phase-corrected by a decision-directed search against the candidate’s constellation, then each symbol is assigned to its nearest ideal point. The Error Vector Magnitude (EVM) of the fit is

$$\text{EVM} = \frac{1}{d_{\min}} \sqrt{\frac{1}{N} \sum_{i=1}^N \|y_i - \hat{c}(y_i)\|^2} \quad (1)$$

where y_i is the i -th received symbol, $\hat{c}(y_i)$ its nearest ideal constellation point, and $d_{\min}$ the constellation’s own minimum point spacing (so a denser grid, e.g. 16-QAM, is not unfairly rewarded just because its points sit closer together).

5.2 Evidence scoring

Every candidate is judged by the same weighted combination of three metrics, each in $[0, 1]$:

$$S = 0.8(1 - \min(\text{EVM}, 1)) + 0.1\mathbb{I}[\text{k-means order matches}] + 0.1 f_{\text{timing}} \quad (2)$$

where the first term (*constellation fit*) dominates and is the discriminating metric; the second (*order consistency*) is 1 only if an independent k-means clustering of the symbols finds the candidate's own order, acting purely as a tie-breaker since k-means can partition any data into any number of clusters; and the third (*timing fit*, f_{timing}) is the strength of the symbol-clock spectral line above the noise expectation, shared by every candidate in a given pass so it scales scores without changing their relative ranking. The weights were fixed empirically and are protected by a regression test on measured classification accuracy.

5.3 SNR estimation (M2M4 moments)

For a complex baseband signal $x = s + n$ (signal plus circularly-symmetric Gaussian noise), let $p = |x|^2$ and define the second and fourth raw moments $M_2 = E[p]$, $M_4 = E[p^2]$. For a noise-free envelope kurtosis $\kappa_a = E|s|^4 / (E|s|^2)^2$ (1 for constant-envelope PSK/FSK, 1.32 for square 16-QAM), it can be shown that

$$M_2 = S + N, \quad M_4 = \kappa_a S^2 + 4SN + 2N^2 \quad (3)$$

where S , N are the true signal and noise powers. Substituting $N = M_2 - S$ gives a closed form for the signal power,

$$S = \sqrt{\frac{M_4 - 2M_2^2}{\kappa_a - 2}}, \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{S}{M_2 - S} \right) \quad (4)$$

clamped to $[-10, +40]$ dB, since moment estimators cannot resolve noise power far below the sampling error of M_4 . κ_a is the hypothesis-specific value, so the reported SNR depends on which modulation is being evaluated.

5.4 Soft-decision bits (log-likelihood ratios)

Rather than discard the demodulator's own confidence, every PSK/QAM/FSK demodulator emits a per-bit log-likelihood ratio alongside its hard decision, using the max-log approximation and the demodulator's own convention that a positive LLR favours bit = 1:

$$L(b) = \frac{d_0^2 - d_1^2}{2\sigma^2}, \quad \sigma^2 = \frac{1}{2N} \sum_{i=1}^N \|y_i - \hat{c}(y_i)\|^2 \quad (5)$$

where d_0^2 , d_1^2 are the minimum squared distances from the received symbol to any constellation point whose relevant bit is 0 or 1 respectively, and σ^2 is the same per-symbol residual already used for EVM (Equation ??), so there is exactly one noise measurement per candidate rather than two that could disagree. This LLR is threaded unchanged through de-interleaving (a fixed bit permutation) into the FEC decoder.

5.5 Soft-decision Viterbi decoding and its significance test

The soft Viterbi branch metric for a trellis transition with expected coded bits $e \in \{0, 1\}^2$ and received LLRs L is the correlation cost

$$\text{bm}(e, L) = \sum_{k=1}^2 L_k (1 - 2e_k) \quad (6)$$

so the minimum-cost path is the maximum-likelihood one. A decode is only reported as successful if its average post-decode Hamming mismatch $\bar{h}$ is more than $z = 4$ binomial standard deviations below the chance level $\bar{h}_0$ measured on a seeded random stream of the same length:

$$\bar{h} < \bar{h}_0 - z \sigma_{\text{pair}}, \quad \sigma_{\text{pair}} = 2\sqrt{\frac{q(1-q)}{2n_{\text{pairs}}}}, \quad q = \frac{\bar{h}_0}{2} \quad (7)$$

which is what prevents the trellis (which fits any bit stream to *some* nonzero mismatch by construction) from being reported as a genuine decode of random data.

5.6 Header/sync-word false-alarm test

For a bipolar correlation $c = 1 - 2d_H/L$ against a known pattern of length L , a window with $m = L(1+c)/2$ matching bits has, under the null hypothesis of a random bit stream, probability

$$p_{\text{single}} = P(\text{Bin}(L, \frac{1}{2}) \geq m) \quad (8)$$

Multiplying by the number of windows and patterns actually tried (Bonferroni correction, including both bit polarities to account for the 180-degree phase ambiguity of BPSK/QPSK) gives the false-alarm probability of the best match found. A header is only reported if this combined probability is below $p < 0.01$; short patterns in long streams therefore cannot qualify by chance.

5.7 Verdict decision logic

The verdict is decided by `core/scoring/verdict.py` in one fixed order, shown in Figure ??.

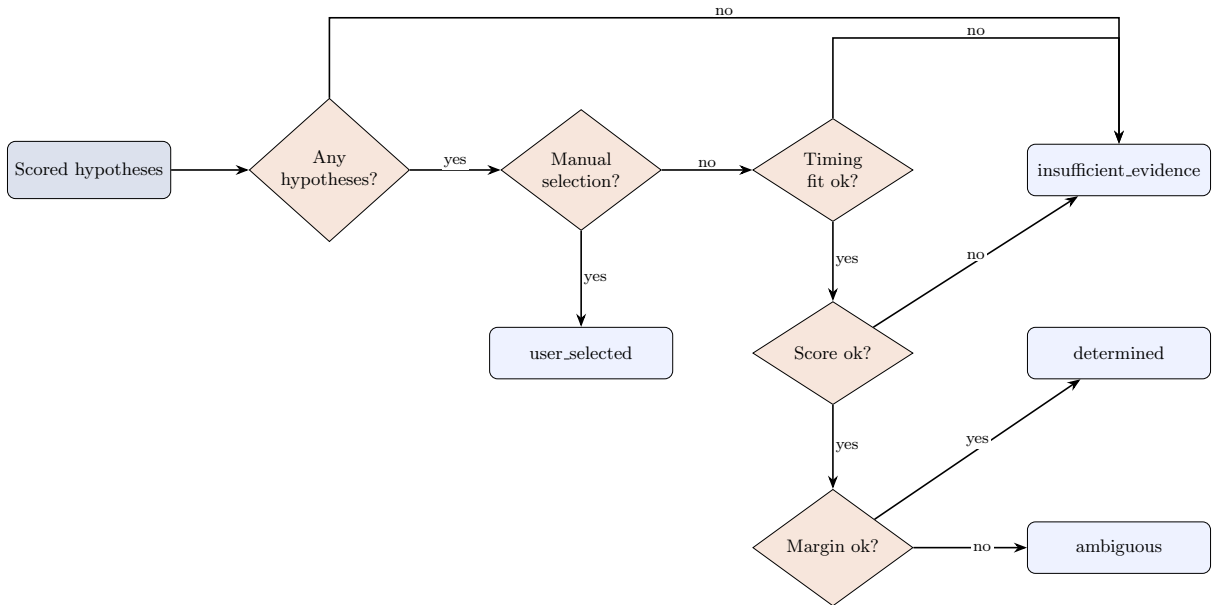


Figure 5: Verdict decision logic, checked in this exact order. “Timing fit ok” means at or above the minimum timing-fit threshold; “Score ok” means the best score is at or above the minimum score; “Margin ok” means the best score beats the runner-up by at least the ambiguity margin.

6 Interface Walkthrough

The screenshots below are the actual, live application (captured via `PrintWindow` on its own window, not mocked up), analyzing `samples/bpsk_carrier_8k.wav`.

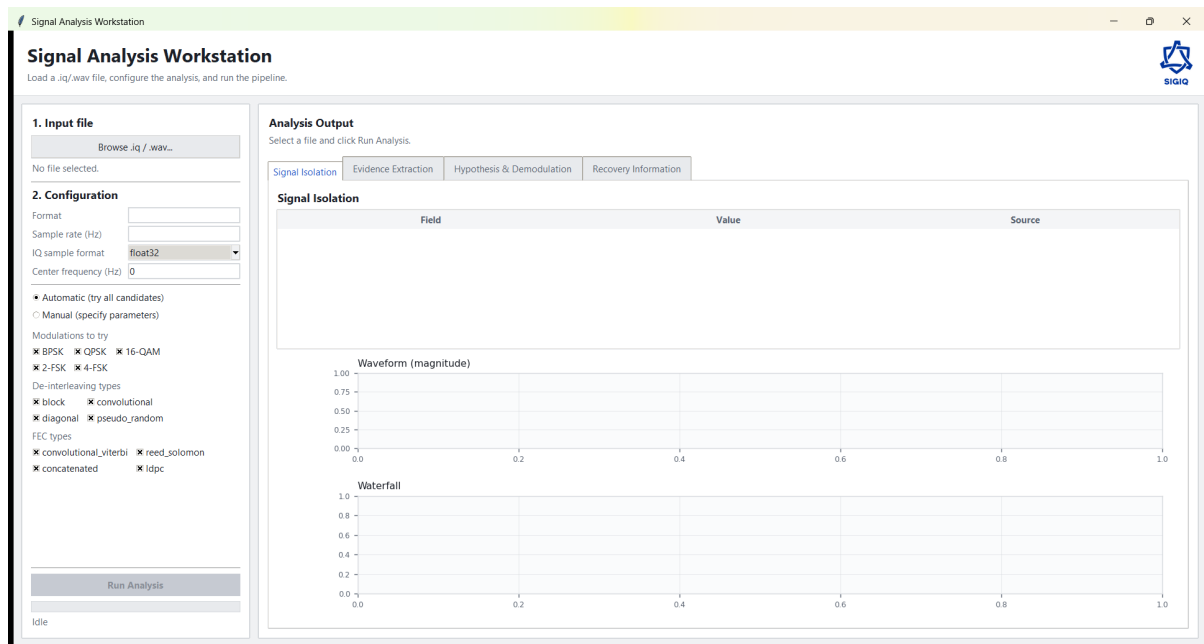


Figure 6: Idle state. The left panel is where every run starts: pick a file, confirm its format details, choose which modulations/de-interleavers/FEC types to try, then click *Run Analysis*.

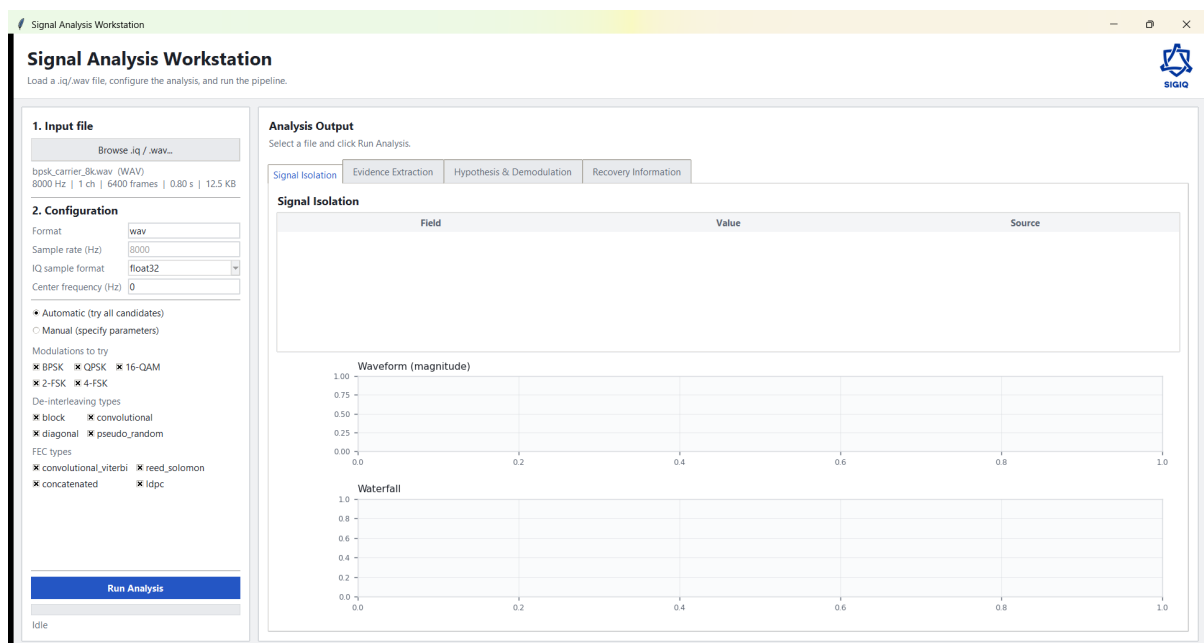


Figure 7: File loaded, ready to run. A *.wav* header supplies sample rate and channel count automatically; a *.iq* file carries none, so sample rate and data type must be entered manually, a real limitation of headerless IQ.

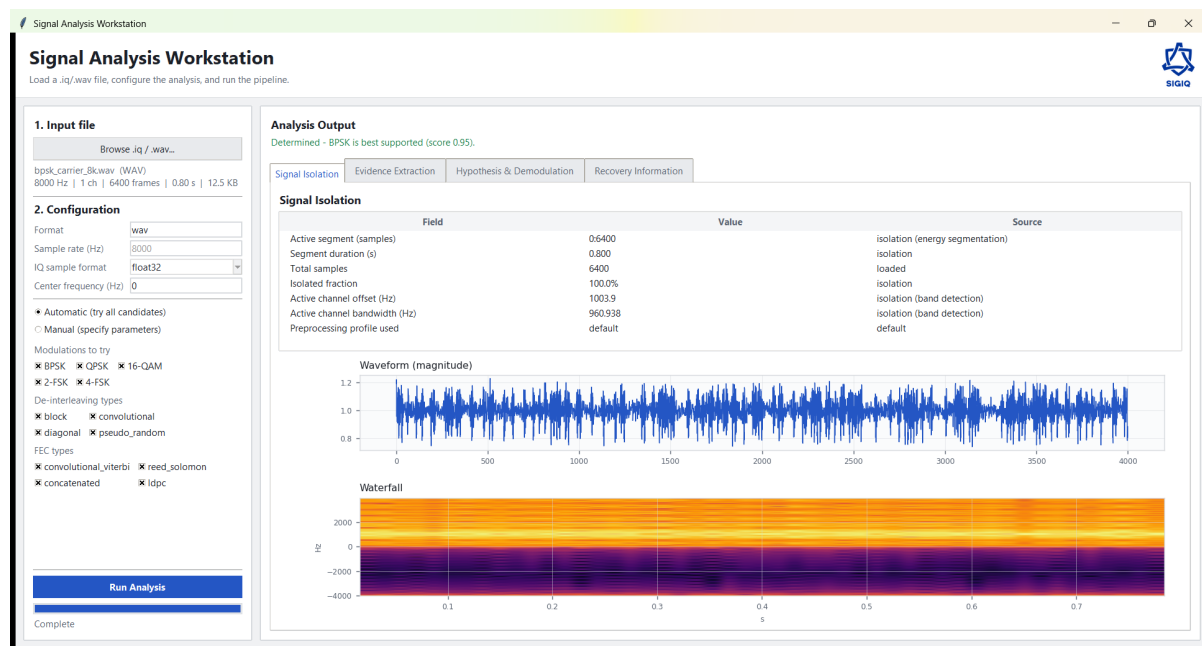


Figure 8: Signal Isolation tab: the active sample range, isolated fraction, detected channel offset/bandwidth and preprocessing profile used, plus the waveform envelope and waterfall of that isolated segment.

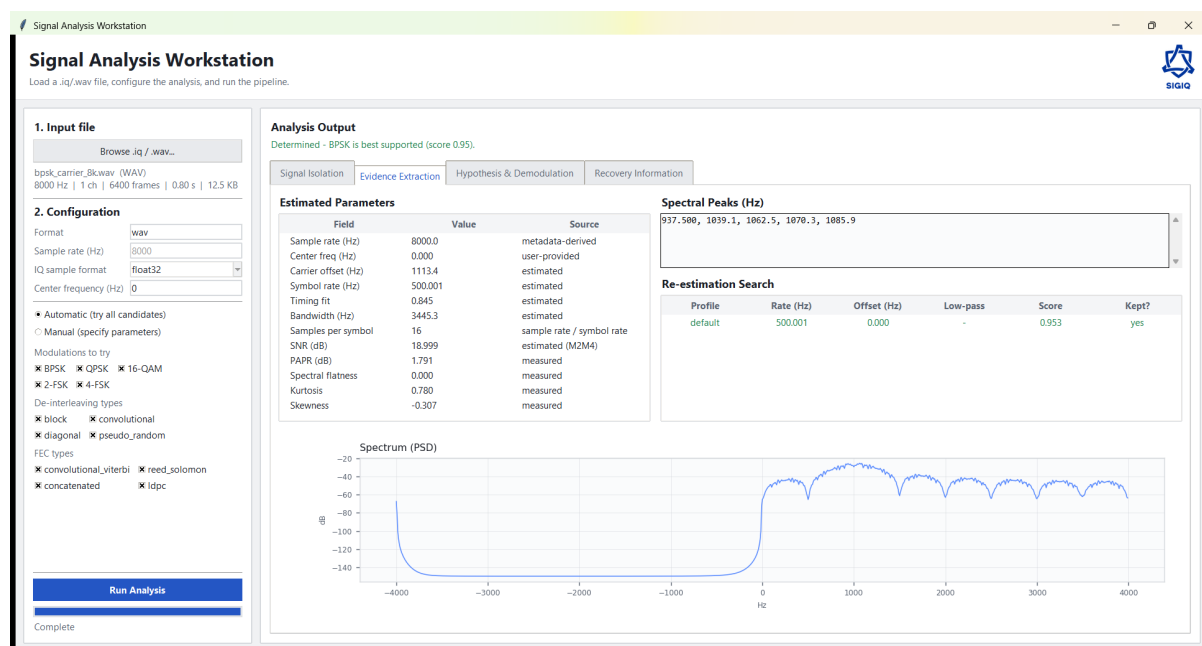


Figure 9: Evidence Extraction tab: estimated parameters each tagged with their source, the re-estimation search trace, and the spectrum (PSD) they were measured from.

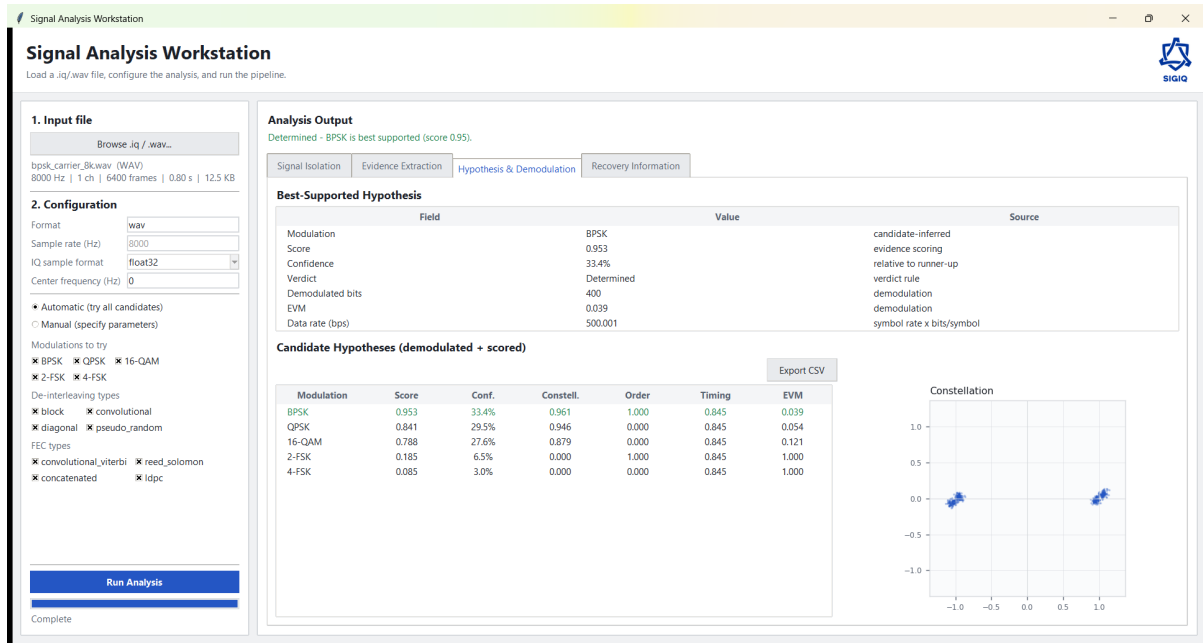


Figure 10: Hypothesis & Demodulation tab: the verdict, every scored candidate with its per-metric breakdown, and the winning hypothesis's own constellation plot.

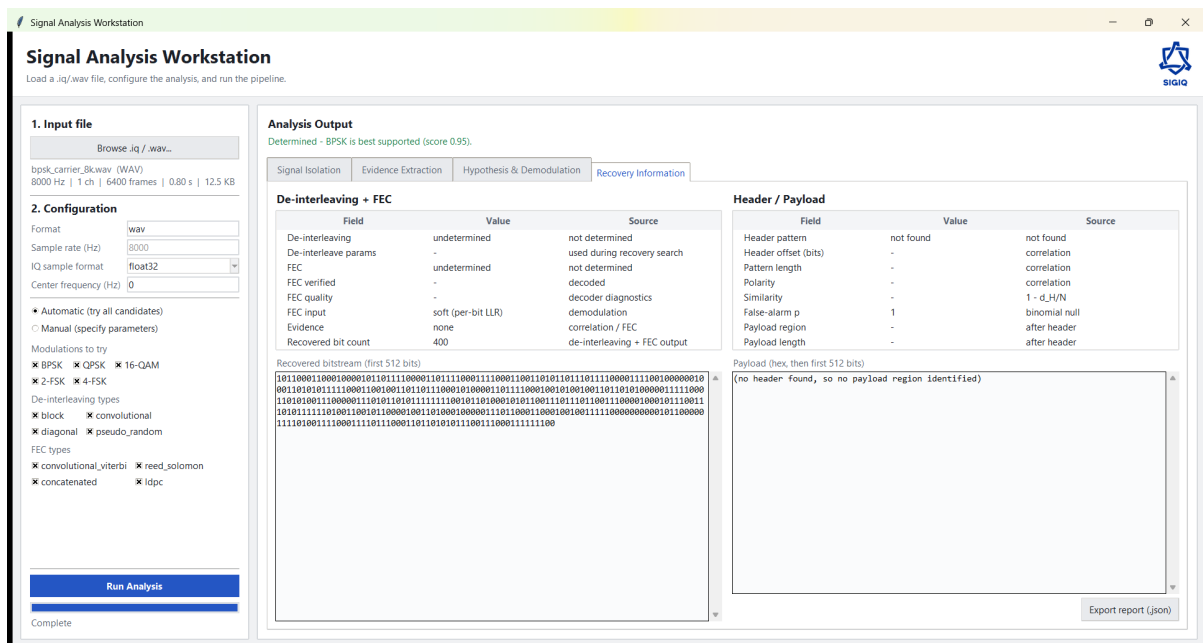


Figure 11: Recovery Information tab: the de-interleaver/FEC choice actually used, decode quality, whether soft LLR information reached the decoder, and header/payload detection with its false-alarm probability.

7 Accuracy and Validation

Measured, not claimed. `tests/test_accuracy_report.py` builds signals with known ground-truth modulation, symbol rate and bits, runs them through the real pipeline, and compares the output numerically, not just whether the best-hypothesis label matches. It is run with fixed, deterministic seeds so results are reproducible run to run, across BPSK/QPSK/16-QAM/2-FSK

at 5/15/25 dB SNR.

Table 3: Measured accuracy, before and after the spec-alignment work documented in this report.

Metric	Result	Starting point
Modulation classification	11/12 (92%)	50%
Mean symbol-rate error	0.0%	4.2%
Mean BER (best-case alignment)	0.046	0.329

The one miss is 16-QAM at 5 dB SNR (measured SNR about 2 dB), which is at the physical limit for this constellation. `tests/test_behaviour.py` separately covers the honesty rules and one end-to-end recovery (raw IQ of coded, framed BPSK, with the interleaver and FEC inferred, recovered exactly): pure noise is not classified, coded streams are verified and random bits are not, headers need statistical significance, manual choices are labelled user-provided, and the re-estimation search stops on tolerance and iteration limit.

8 Known Limitations

Documented, not hidden.

Area	Limitation
Classification	16-QAM at very low SNR is misclassified as QPSK; 2-FSK whose tone spacing equals the symbol rate is spectrally hard to separate from BPSK.
FSK detection	Low modulation-index FSK (tone spacing a small fraction of the symbol rate) is not detected at all: the PSD is a single unimodal lobe with no resolvable per-tone humps, and no threshold-based fix was found that did not also false-positive on PSK/QAM.
SNR estimation	The M2M4 estimator (Equation ??) saturates near the +40 dB clamp for true SNR above roughly 20–25 dB, a finite-sample instability in the fourth-moment estimate at high SNR.
Symbol-rate estimation	Can still lock onto an exact $2\times$ harmonic for unshaped (rectangular-pulse) PSK/QAM during a second preprocessing-profile retry, because sampling at an exact multiple of an unshaped signal’s true rate just repeats each real symbol; EVM cannot tell that apart from the true rate.
LDPC	Supports exactly one code, the IEEE 802.11n $n = 648$ rate-3/4 code, with min-sum decoding. Other LDPC codes are reported as not decodable.
Reed-Solomon	Hard-decision only; classical Berlekamp-Massey decoding has no soft form.
Header detection	Knows three sync words (CCSDS ASM, HDLC flag, Barker-11). Short patterns in long streams cannot pass the false-alarm test.

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Area	Limitation
Verdict logic	Does not yet distinguish “wrong because the constellation is not supported” from “wrong with real ambiguity”; an unsupported order like 64-QAM can come back determined with an incorrect modulation.
Timing recovery	Decimates at an integer samples-per-symbol phase; no fractional-rate interpolation.

9 Conclusion

SigIQ combines automated modulation identification, parameter estimation and signal recovery inside a single desktop application, built against one governing rule: never report a result the evidence does not support. Every candidate is genuinely demodulated and scored rather than guessed; every decoded, header-found or high-confidence result is backed by an explicit statistical test against chance; and honest **ambiguous** and **insufficient_evidence** verdicts are treated as correct outcomes, not failures to be hidden. Measured against ground-truth synthetic signals, the system classifies modulation correctly 92% of the time with 0.0% mean symbol-rate error, up from a 50%/4.2% starting point before the evidence-scoring and re-estimation work documented in this report. The known limitations in Section ?? (low modulation-index FSK detection, SNR saturation at high true SNR, and the residual symbol-rate harmonic-lock path) are recorded as concrete, reproducible directions for future work rather than left undiscovered.